Problem 8: Duty Roster Rotation

TASK

A hostel rotates its cleaning duty roster every week by moving each name k places to the right; names that go past the end wrap around to the front.

Implement a function rotateRoster(names, k) that returns the roster after rotating it to the right by k places. k may be larger than the number of names.

CONSTRAINTS

- 1 <= length of names <= 100
- 0 <= k <= 10^9

EXAMPLES

Sample 1

names = [A, B, C, D, E], k = 2

Expected Output: [D, E, A, B, C]

Sample 2

names = [A, B, C, D, E], k = 7

Expected Output: [D, E, A, B, C] // 7 places is the same as 2

COMPLEXITYREQUIREMENTS

State the time and space complexity of your solution. Explain why rotating one place at a time, k times, is slow when k is large.

HINT

Use k % n to remove full turns. Each name at index i moves to index (i + k) % n.